RES-ME 5704: Introduction to Biostatistics

Lecture 3: Variables, Descriptive Statistics, and Data Visualization

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Spring 2023 Week 3

Outline

1. Understanding measurements
   - Data Types
   - Exploratory Data Analysis
2. EDA: Graphs
   - Distributions
     - Common Distribution Graphs
     - For Categorical Data
     - For Continuous Data
3. EDA: Statistics
   - Percentiles
   - Measures of Center and Spread
4. References

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Data Types

Data Types

All statistical methods have assumptions.

- For almost all of them you need to know the type of data that you have.

- We will look primarily at 2 types:
  1. **Qualitative (Categorical)** Data
  2. **Quantitative (Numerical)** Data



Data Types

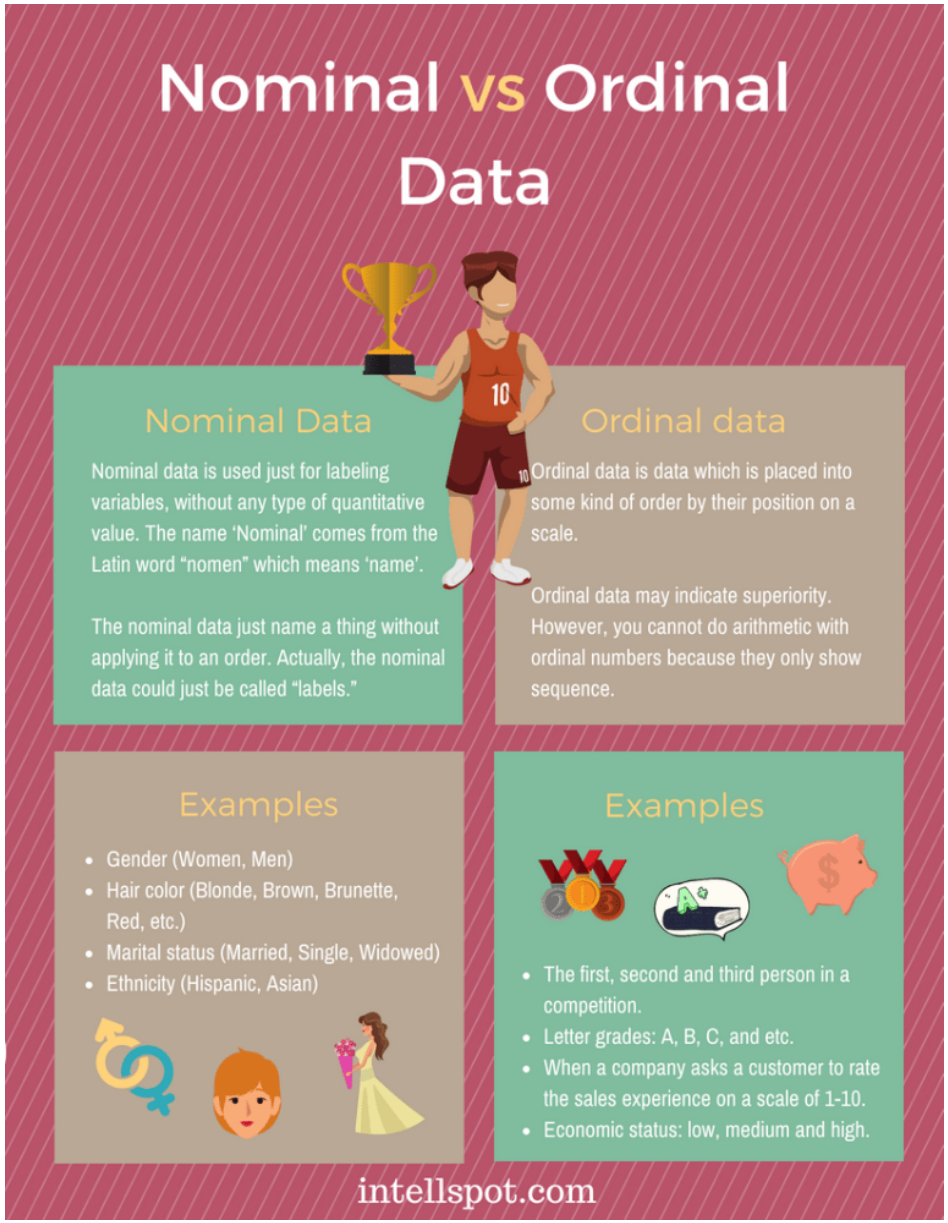
Categorical (or Qualitative) Data

Think of this as data that count things having a given **quality**.

- Characteristics that are **categorical** (male vs. female). The possible categorical values are called **categories or levels**.

Often the levels of the variables are mutually exclusive (e.g. State of residence), but not all categorical variables are that way (e.g. Race/Ethnicity levels).

Data Types



Categorical or Qualitative Data

- **Nominal**: Names
  - Race: White; African American; Asian; Hispanic
- **Ordinal**: Rank-order
  - Amount: low/medium/high
- **Binary** - Can be nominal or ordinal
  - 2 categories: yes/no; male/female; low/high

Data Types

Numerical (or Quantitative) Data

Think of this as data that measures **quantities**.

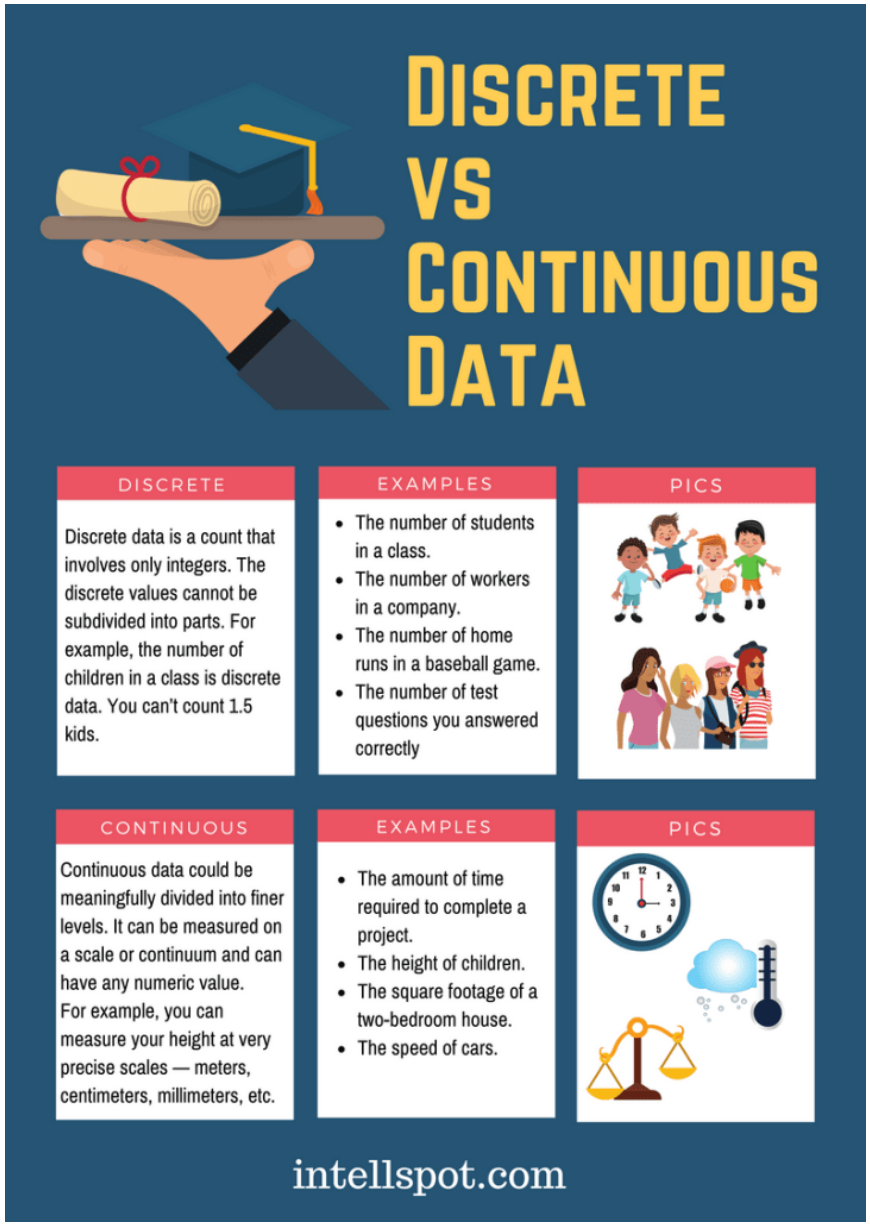
- This type of data can take on a wide range of numerical values and it is sensible to add, subtract, or take averages with those values.

- Not all data with numbers would be considered numerical! E.g. a variable reporting phone number area codes or an identification variable.

Data Types

Continuous or Quantitative Data

- **Discrete or Count**
  - number of siblings or dmft
- **Continuous**
  - can take on any value depending on how measured: Blood pressure or age



Class Discussion: Data Types

Class Discussion: We will watch the following YouTube Video tutorial and work together to answer the tutorial questions.

Types of data - practice identifying, Shopping Mall example (4:22)

Exploratory Data Analysis

EDA - Exploratory Data Analysis

Exploratory Data Analysis

- Descriptive statistics (means, medians, etc.)
- Graphical displays (stem and leaf plots, boxplots, etc.)

ALWAYS start any data analysis by exploring your data

- Summarizes data into manageable bits
- Helps discover errors
- Helps you learn what is going on in your dataset

Describing your data

It is imperative that you describe your data in a way that can easily summarize large amounts of information in a concise way.

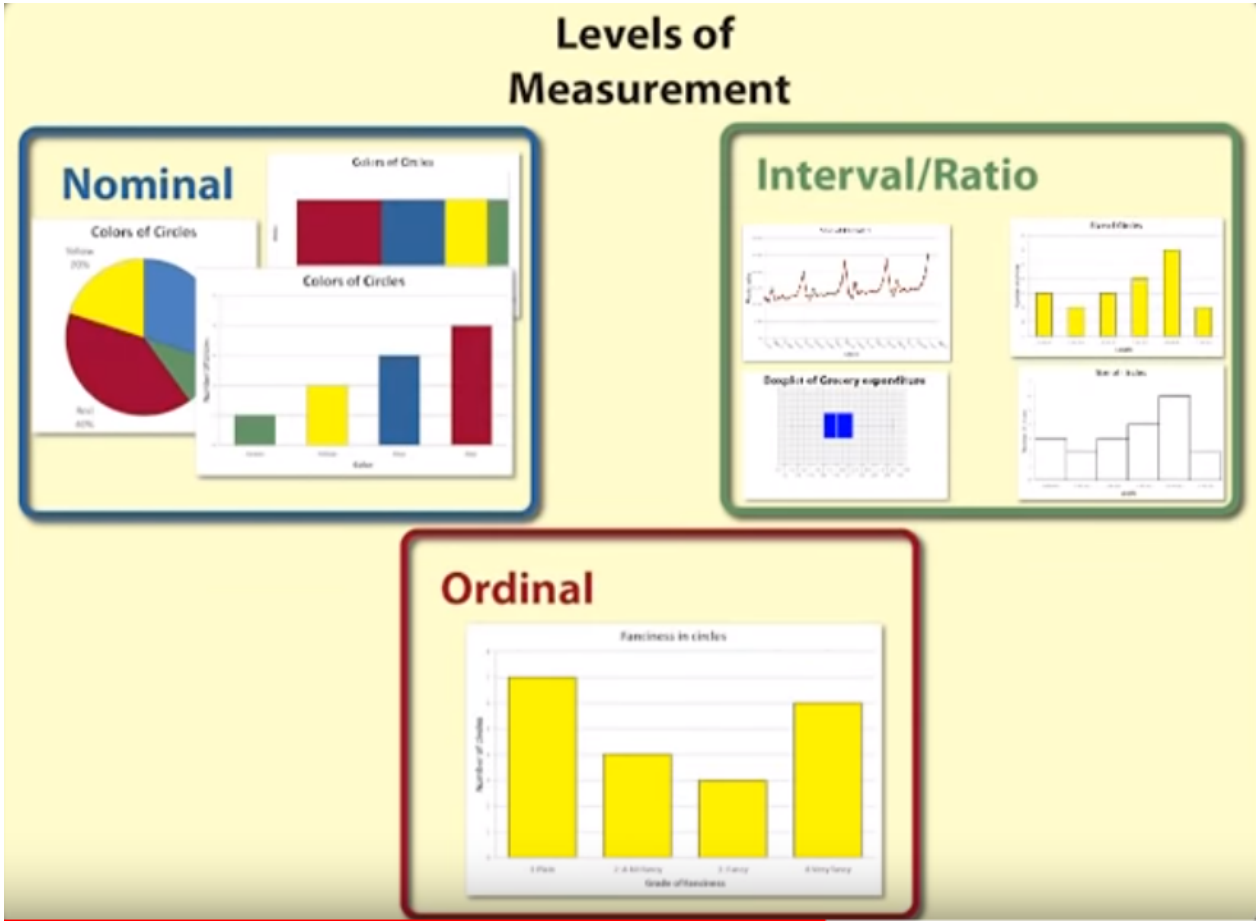
There are two main ways you can describe your data:

1. With **numbers** or
2. with **pictures**

It is common to start describing your data by describing each variable **numerically**.

- This is called **Univariate Analysis**.
- However, you can describe your data both numerically or pictorially at the same time (on certain types of data).

Describing your data: **Pictorially**



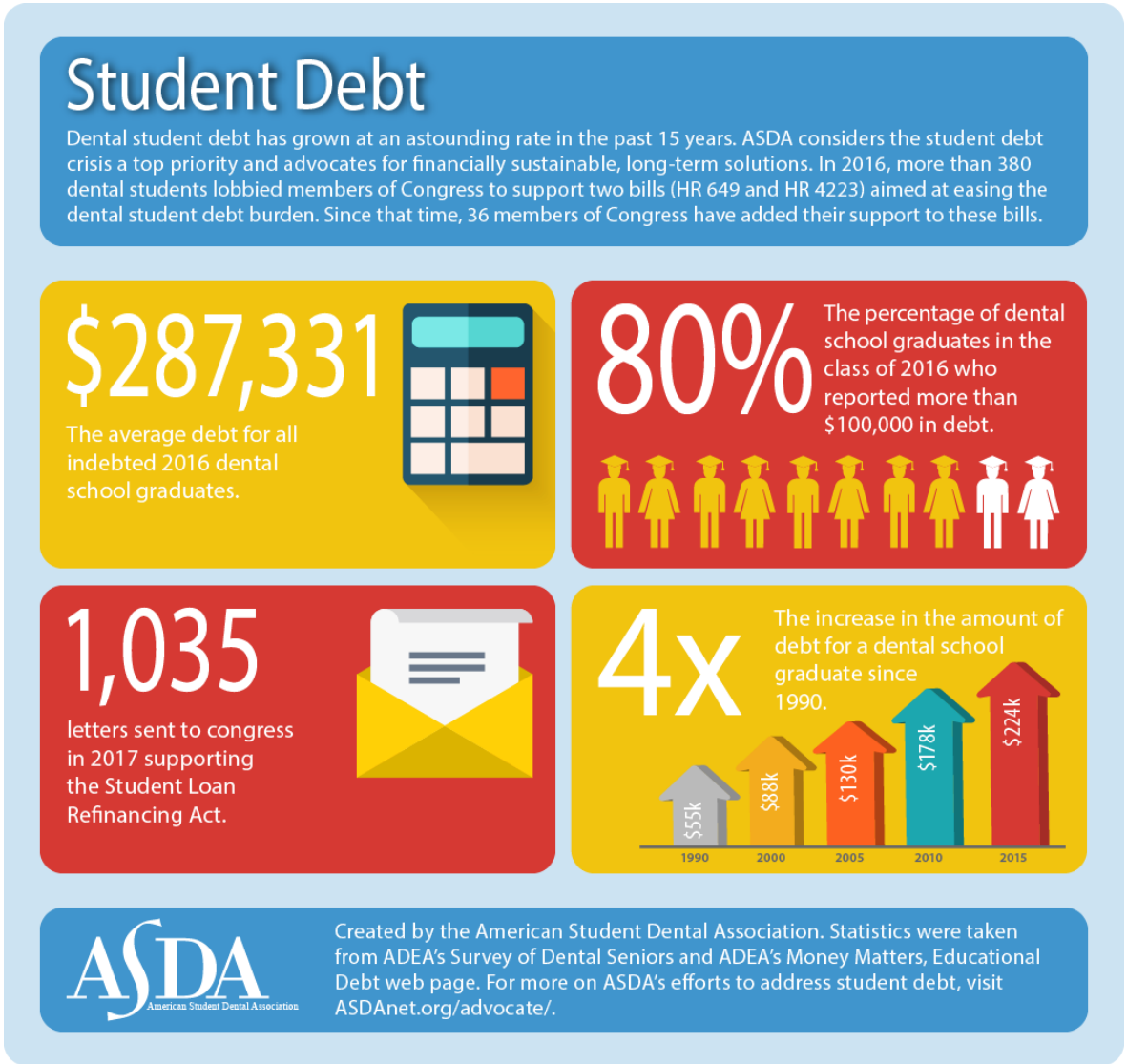
The graphs (pics) you use depend on the type of data you have!

- **Continuous data**: Histograms (bar charts) or boxplots
- **Categorical data**: Column (or bar) charts or pie charts (except ordinal data)

Describing your data: **Numerically**

Depends on the type of data you have!

1. **Continuous data**
   - *Measures of location*: The 'typical' value (center)
   - *Measures of spread*: How close an individual value is to that 'typical' value
2. **Categorical data**
   - Counts and percentages



Outline

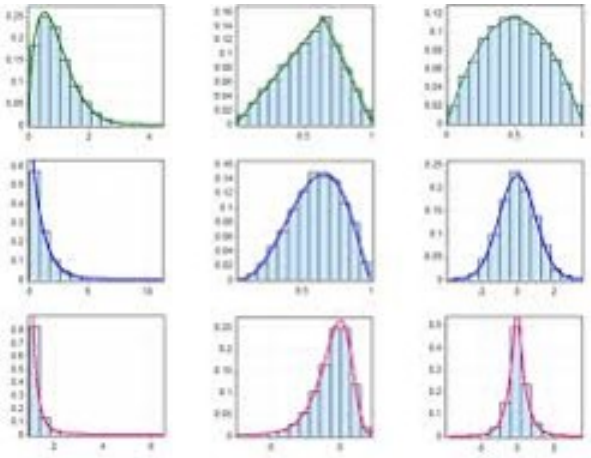
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Distributions

Distribution

An additional assumption for most statistical tests has to do with the **distribution of the data**.

- I think of the distribution of the data as the data's **shape**.
- Others call it the data density.



Regardless of what you call it, **you need to have an idea of what the data looks like pictorially before you perform any statistical tests**!

Common Distribution Graphs

Types of Graphs

The type of graph you can create depends on **the type of data** you have!

FOUR TYPES OF DATA		TYPES OF CHARTS
NOMINAL: Data sorted into categories	⟶	Bar chart
ORDINAL: Arbitrary numerical scale	⟶	Pie chart, bar chart
DISCRETE: Represents units	⟶	Arrays, Pie chart, Bar chart
CONTINUOUS: Can be measured on a continuum	⟶	Line chart

Frequency Distribution

The most basic tool for presenting data is a **frequency distribution**

- The word *frequency* here refers to a **count** rather than 'happening often in time'.

A **frequency distribution** gives

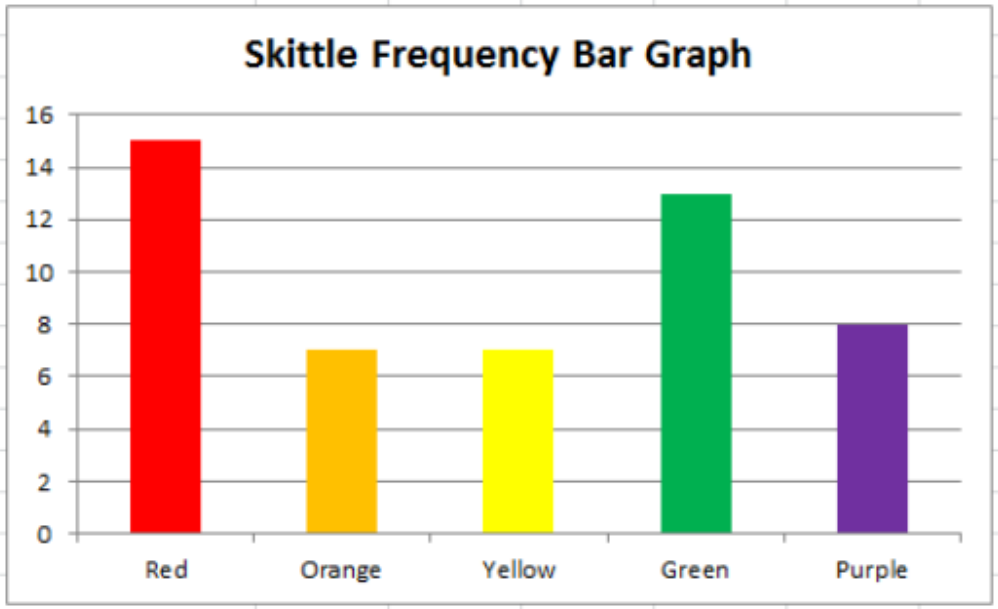
- the count of subjects in each group (for **categorical** data)
- OR in some range of value (for **continuous** data)

Common Distribution Graphs for Categorical Data

Bar or Column Charts

Bar or column charts are a type of graph using to describe **categorical data**, i.e., nominal or ordinal data.

- In most bar charts, you can think of the word *distribution* as referring to the **horizontal axis** . . . to the categorical levels.
- The vertical axis gives the *frequencies* - the count of the subjects in a category.

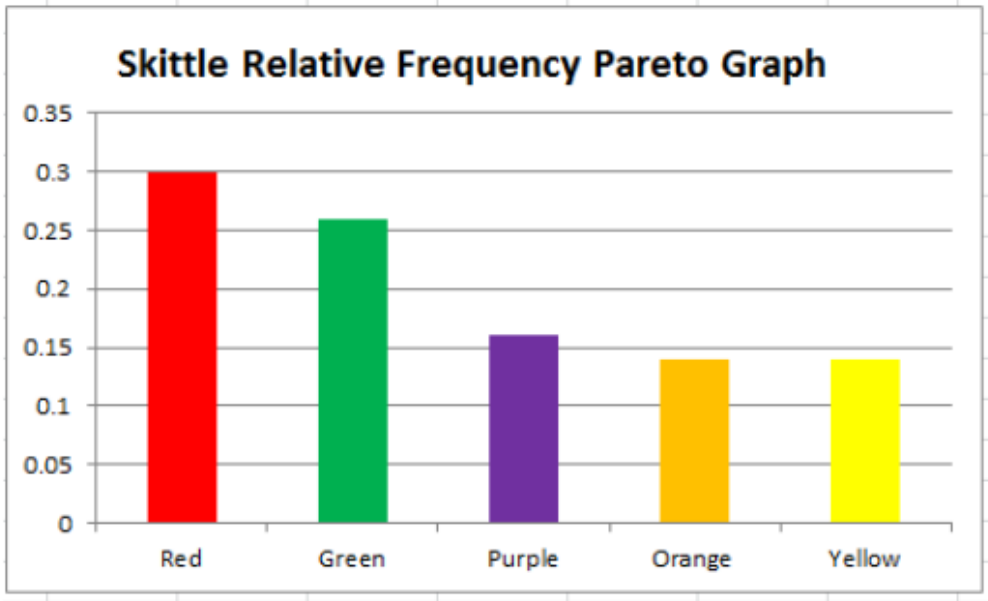


For bar charts of nominal data, the order of the bars does not matter. With ordinal data, you should order the bars so that the inherent order of the data is correct. (Graph Source)

Relative Frequency Bar Charts

Relative Frequency Bar Charts are a type of bar graph that uses the relative frequency instead of the frequency.

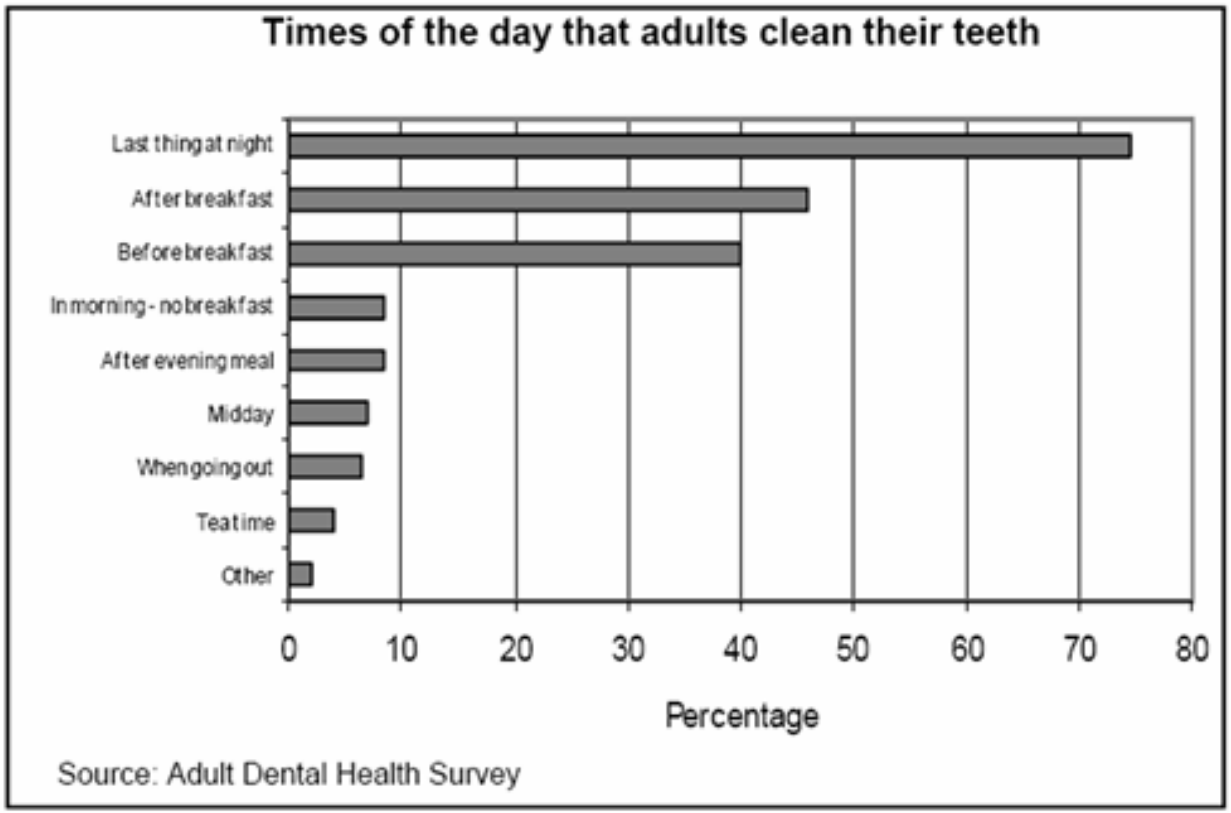
- The **relative frequency** is found by dividing the frequency in each category and dividing by the total frequency in the sample.
- Basically, the percentage of each category.



A special type of bar graph where the bars are drawn in decreasing order of relative frequency is called a **Pareto chart**.
(Graph Source)

Horizontal Bar Charts

Horizontal Bar Charts are simply bar charts with the distribution on the vertical axis and the frequency on the horizontal axis.



For any bar chart, the 'frequency' can be written as a whole number, a decimal, a percentage, or as a fraction.
(Graph Source)

Grouped Bar Charts

Grouped Bar Charts are bar charts the distribution grouped by another variable (subgroups of the main categories).

- A separate bar represents each subgroup (e.g. gender) and are usually colored or shaded differently to distinguish between them. A legend or key is usually provided.

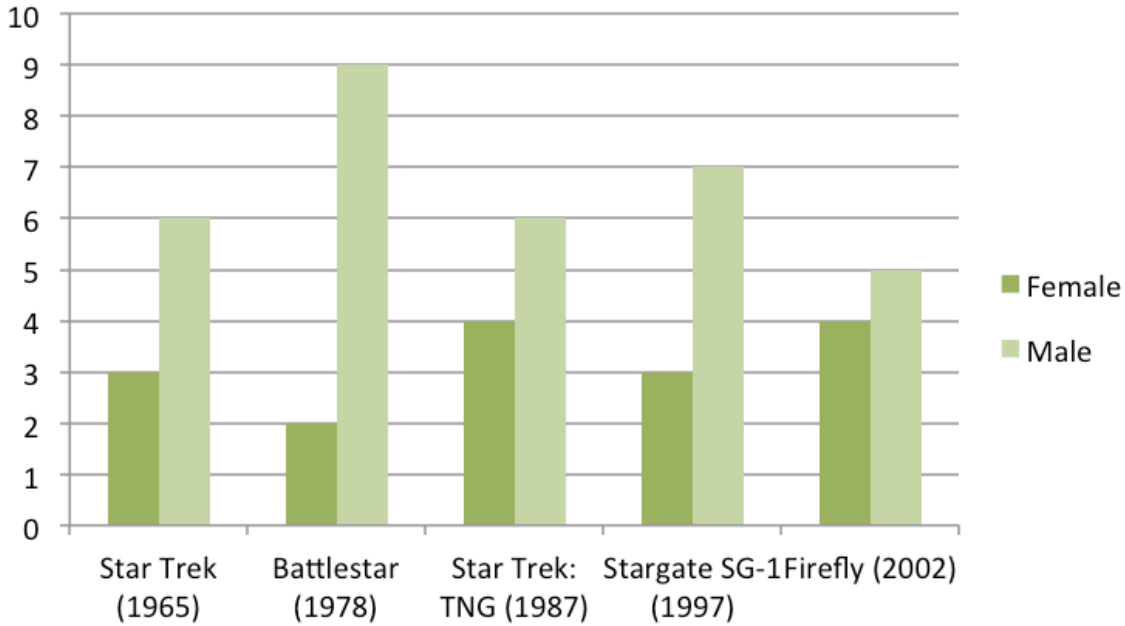


Figure 3. Genders of spaceship crew members in popular television series

Since bar charts involve distinct categories, it's standard to make sure that there are gaps between the bars (categories) of main variable.

(Graph Source)

Grouped Bar Charts



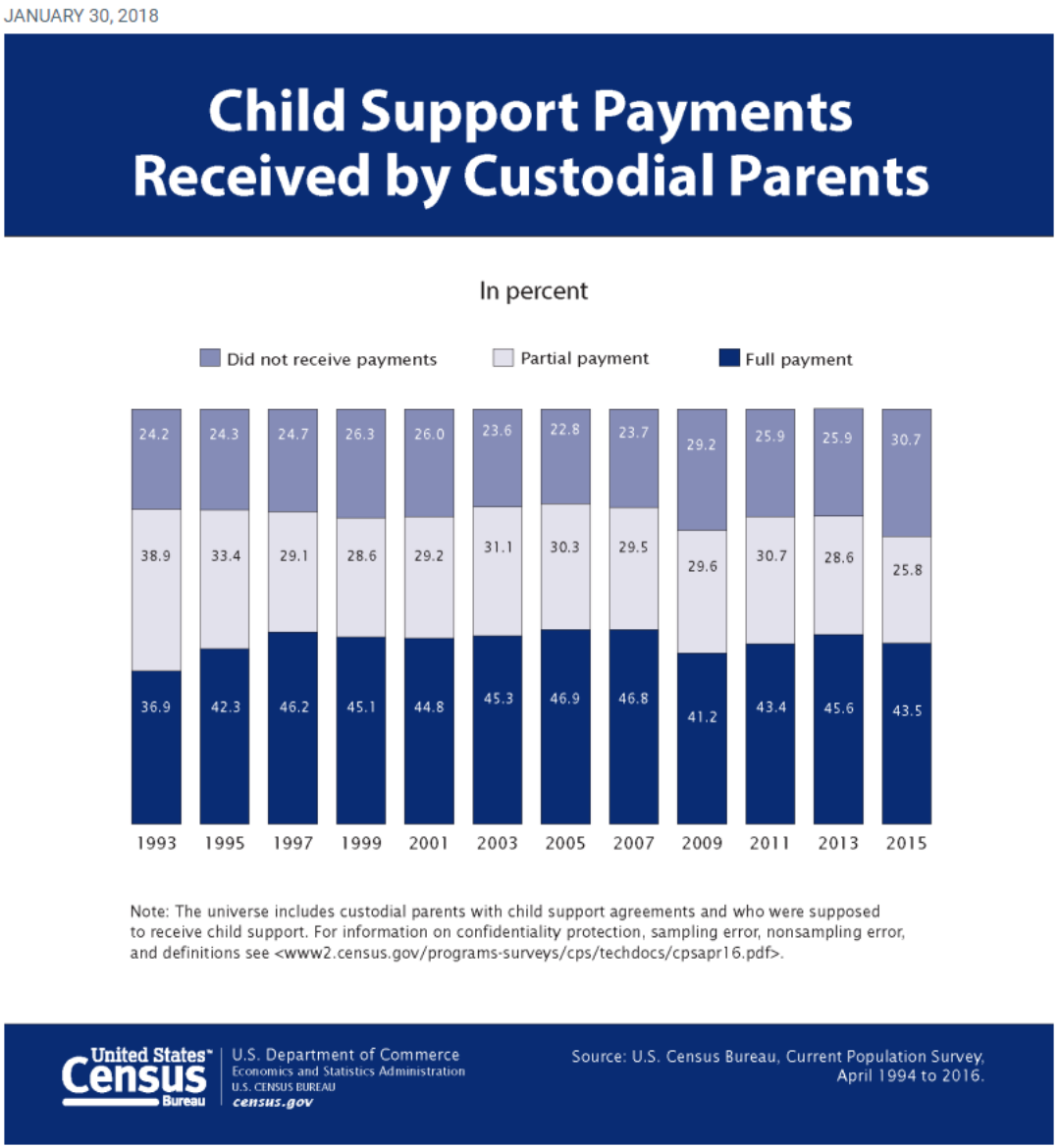
(Graph Source)

Grouped Bar Charts can also be horizontal!

Stacked Bar Charts

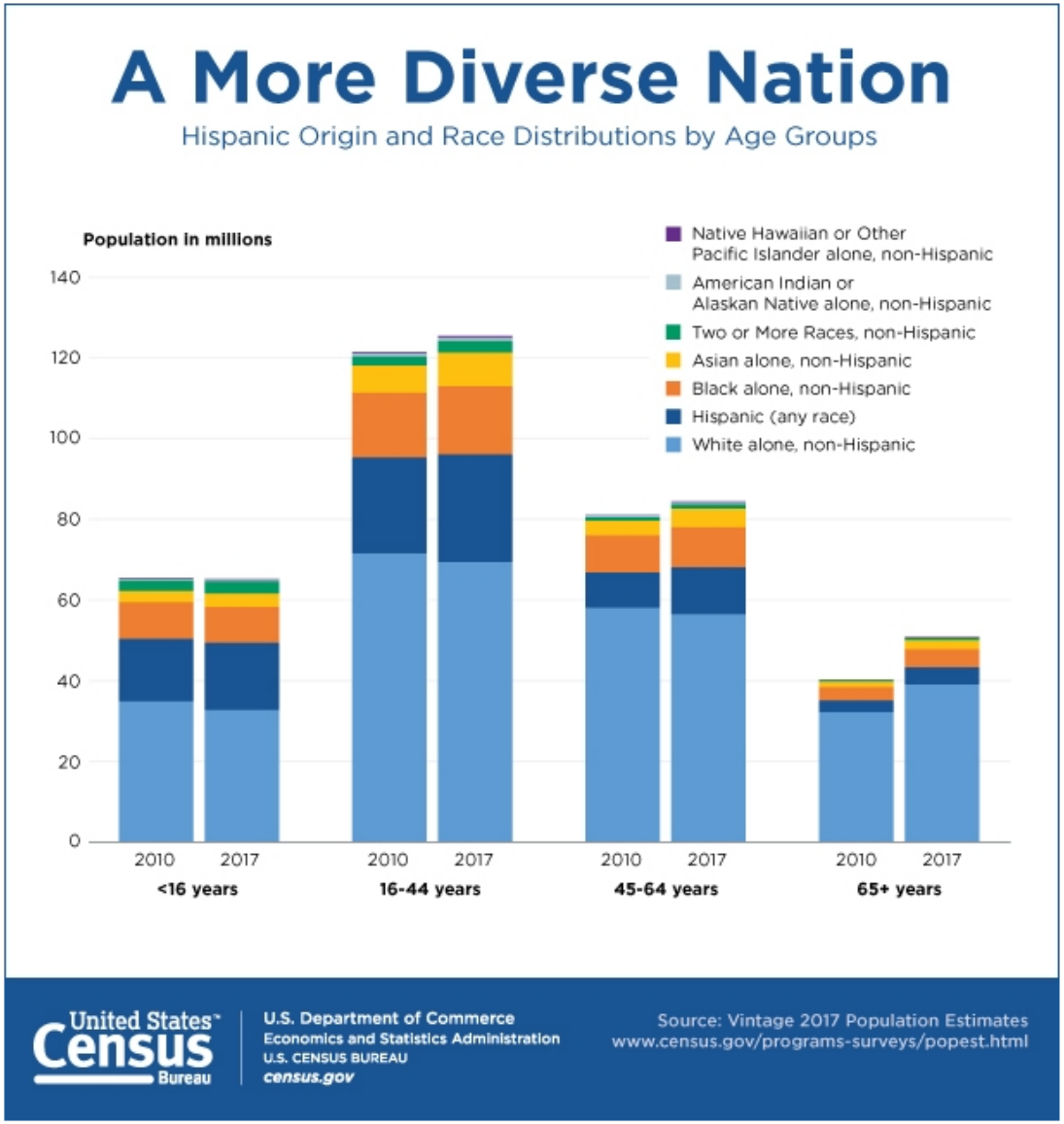
Stacked Bar Charts are similar to grouped bar charts in that they are used to display information about the sub-groups that make up the different categories.

- The bars representing the subgroups are placed on top of each other to make a single column, or side by side to make a single bar.
- The overall height or length of the bar shows the total size of the category; different colors or shadings are used to indicate the different subgroups.



(Graph Source)

Stacked Bar Charts



(Graph Source)

Stacked Bar Charts can also be grouped!

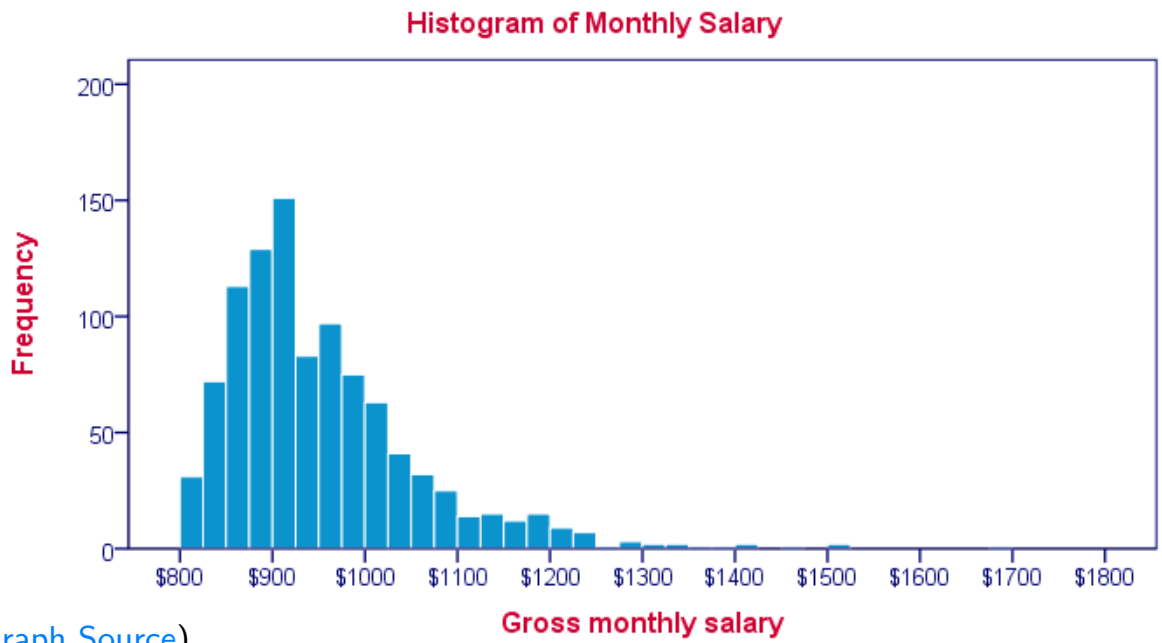
Common Distribution Graphs for Continuous Data

Histograms

Histograms

Quantitative data can be presented in bar graphs in the same ways as qualitative data. A bar graph for quantitative data is called a **histogram**.

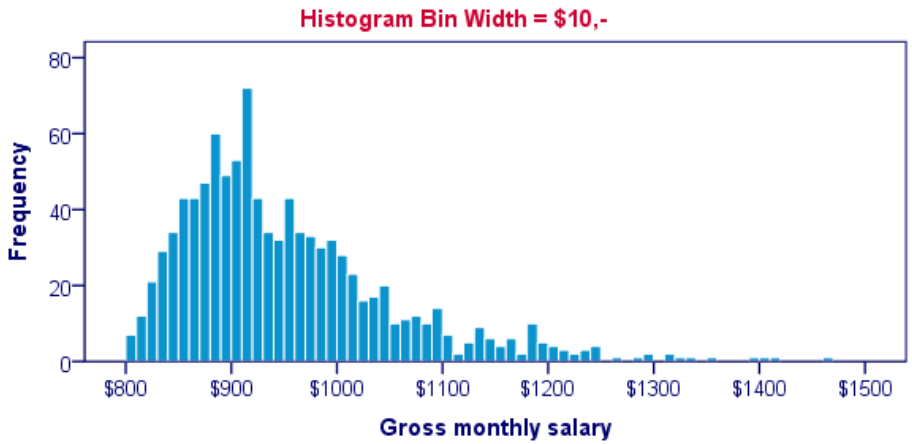
- The values taken by the data are labeled in ascending order across the horizontal axis and the height of each bar corresponds to the variable's frequency or relative frequency.

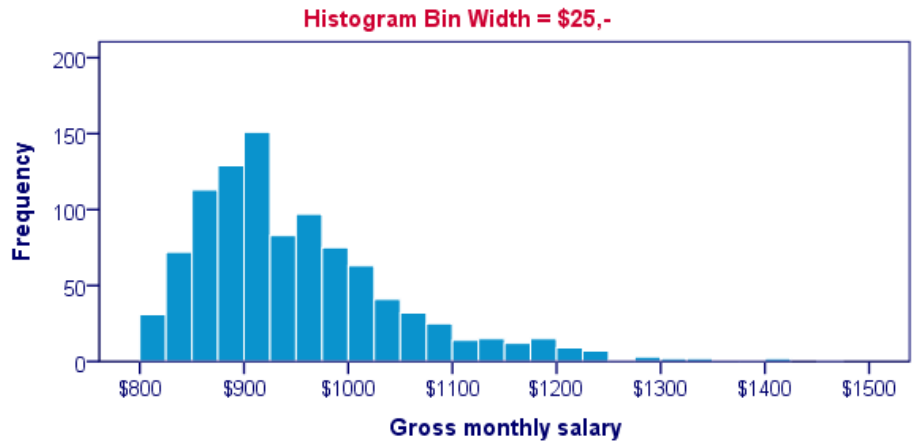


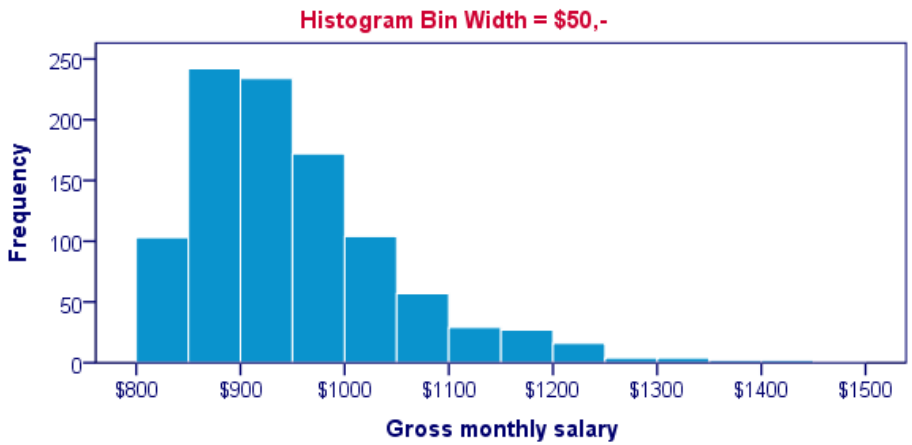
(Graph Source)

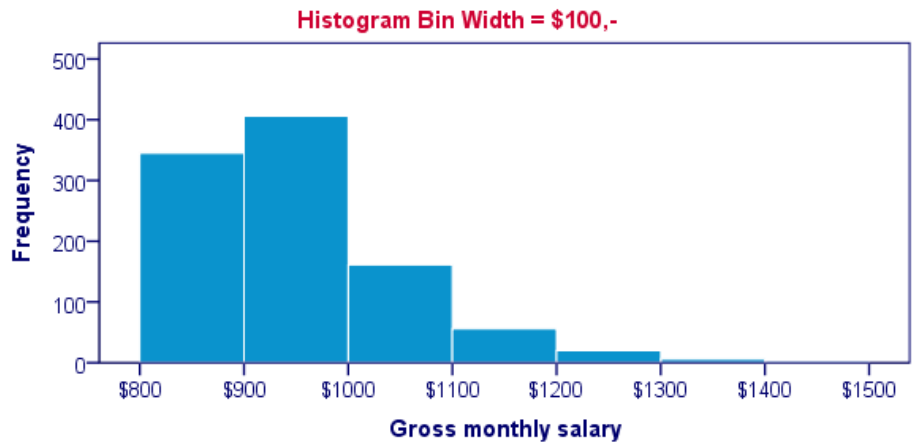
Histograms - Bin Width

The bars in a histogram are called **bins**. And bin size matters!!





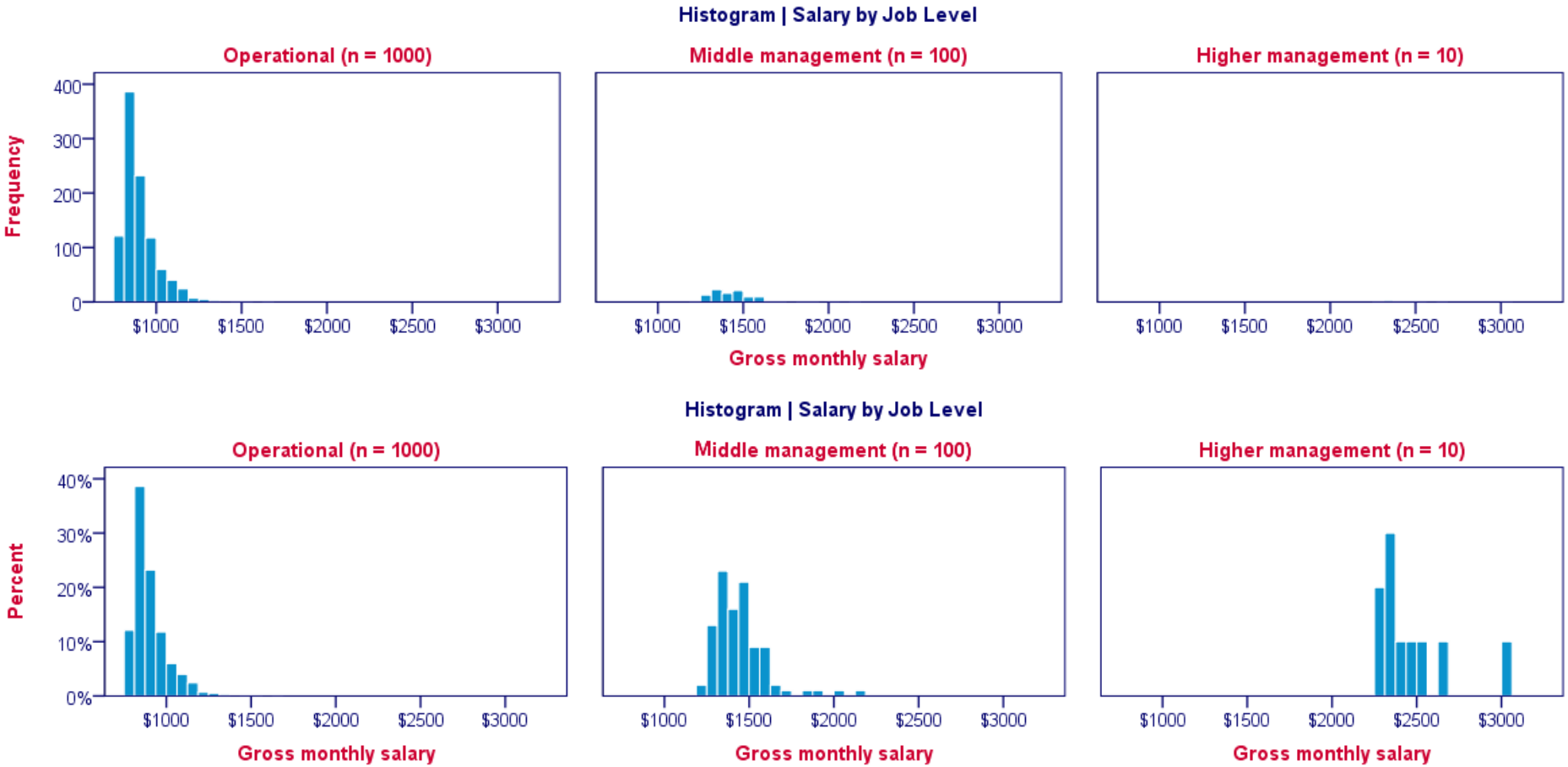




(Graph Source)

Histograms - Frequencies vs Percentages

The axis choice matters too!!

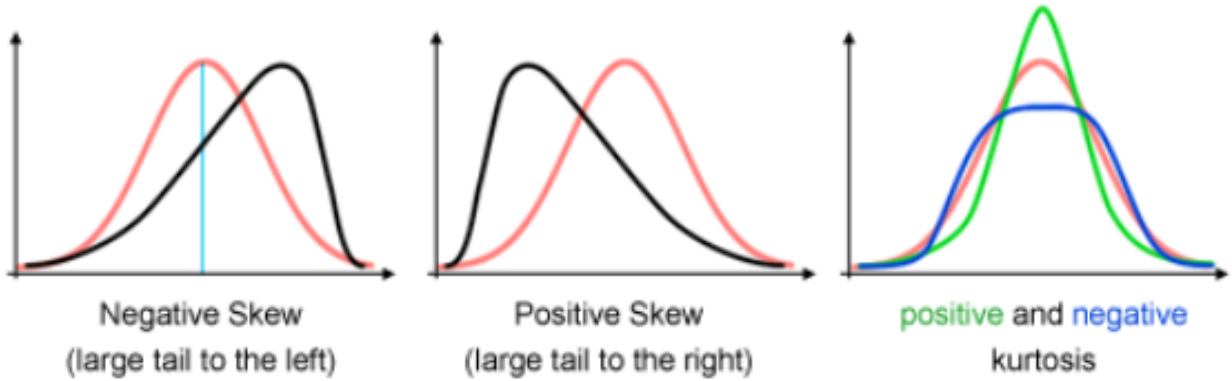


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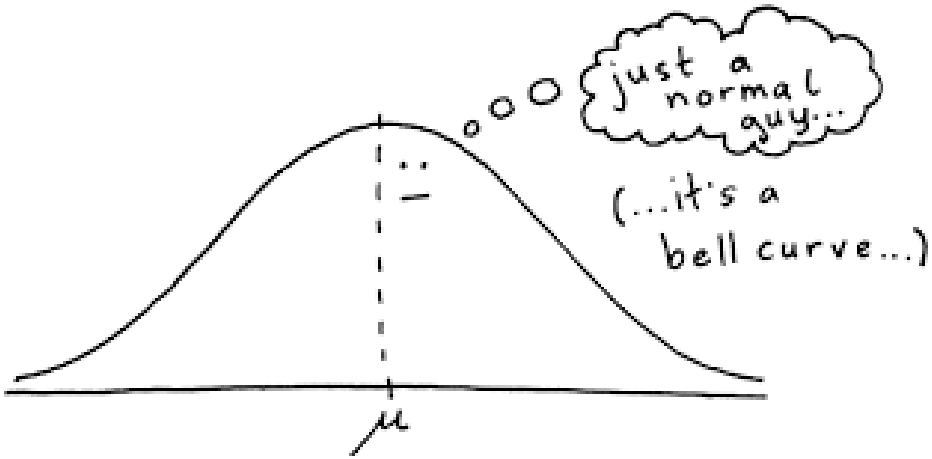
Histograms - Shape

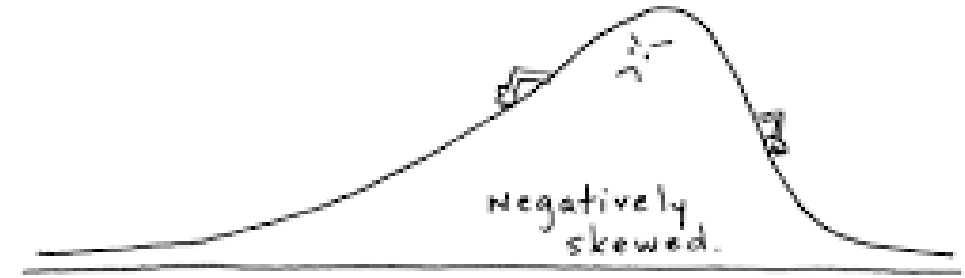
Here are the common distribution shapes we'll encounter:

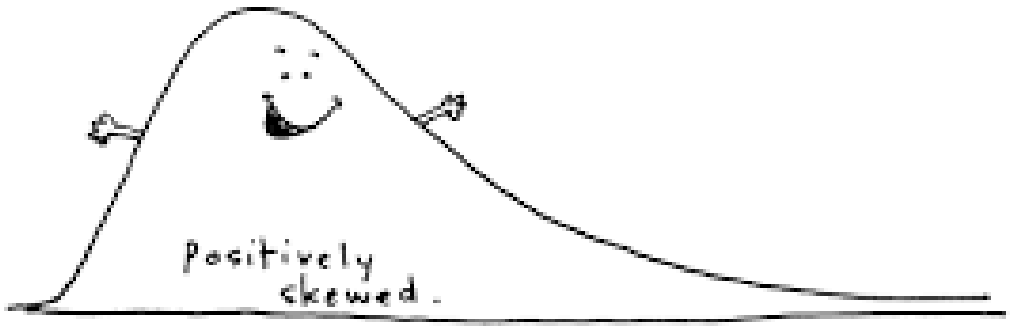
- **Bell-Shaped**: Most of the values fall in the middle (a central peak), and the frequencies tail off to the left and to the right symmetrically.
- **Positive Skewed**: Not symmetric. The tail to the **positive side** is longer.
- **Negative Skewed**: Not symmetric. The tail to the **negative side** is longer.
- **Positive Kurtosis**: Symmetric. The peak is too tall.
- **Negative Kurtosis**: Symmetric. The peak is too flat.



Histograms - Skewness

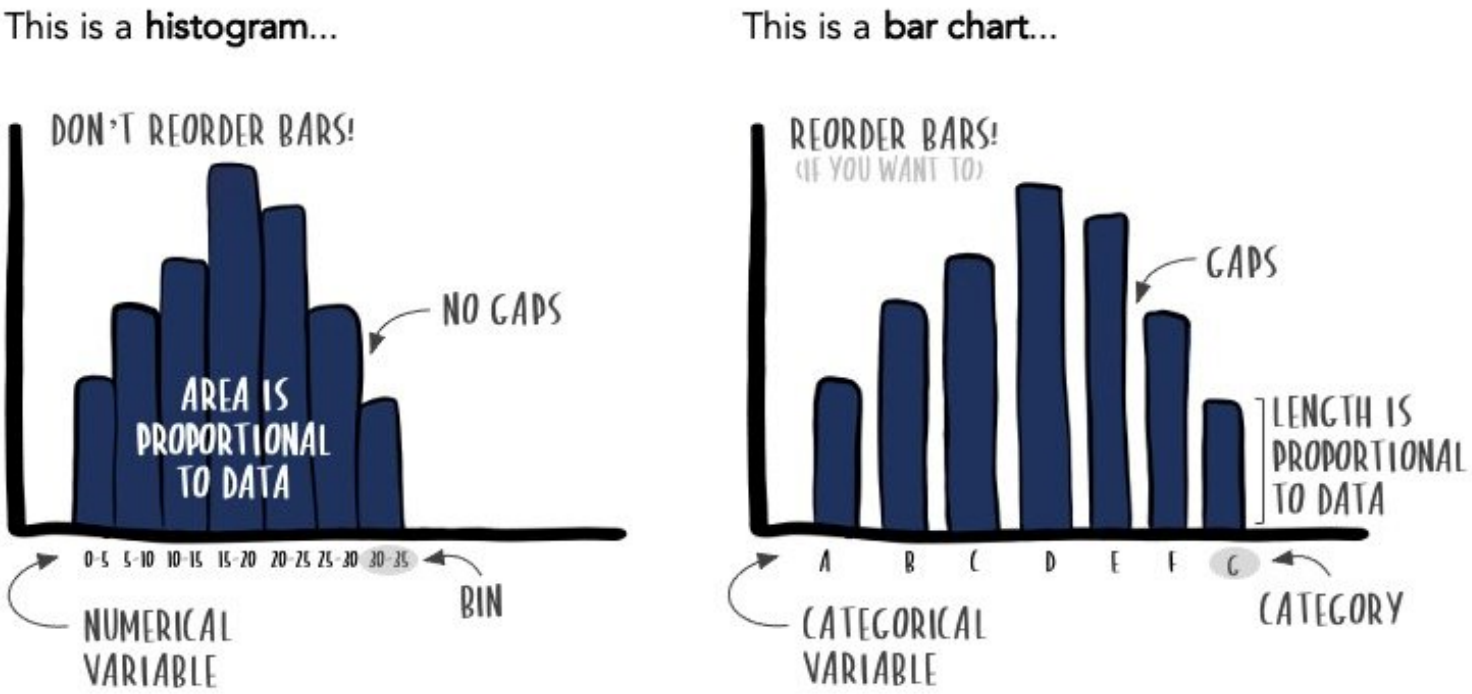






Histograms vs Bar Charts

The main **visual** difference between a bar graph and a histogram is that there should be no horizontal spacing between numerical values along the horizontal axis. In other words, **bars touch each other in a histogram**!



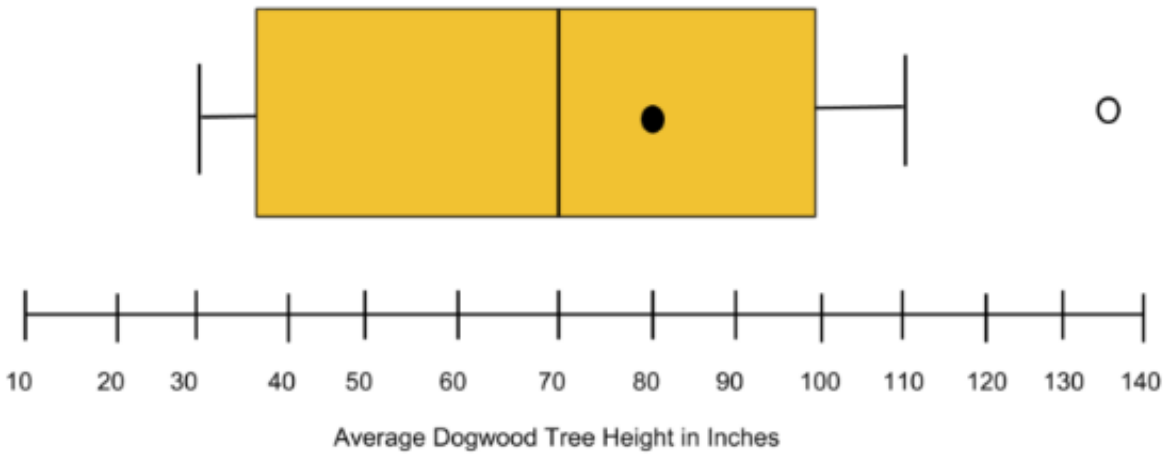
(Graph Source)

Boxplots

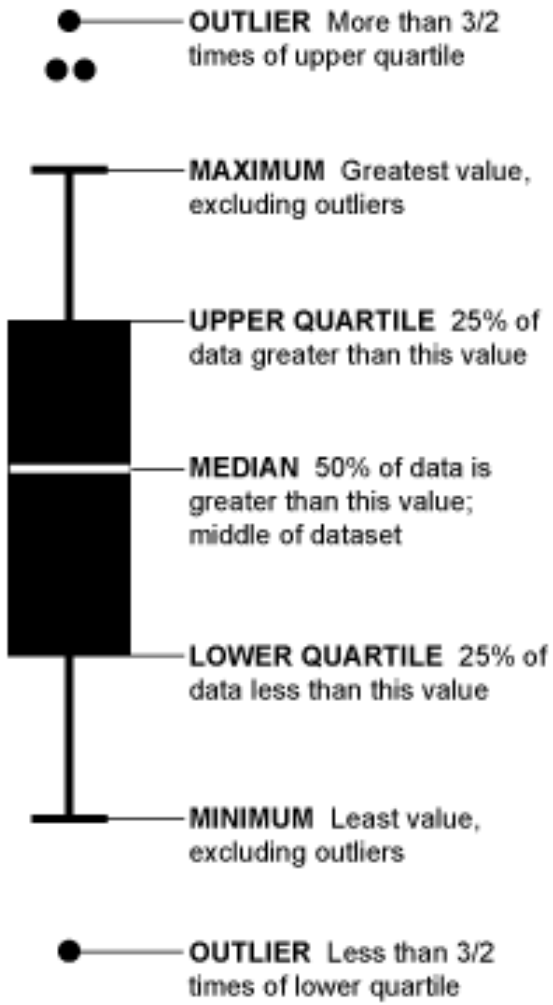
Boxplots

Another way to look at the shape of the data is using a **boxplot**.

You can see the components of a boxplot are very technical. More importantly, you can look at one to see the shape of the distribution.



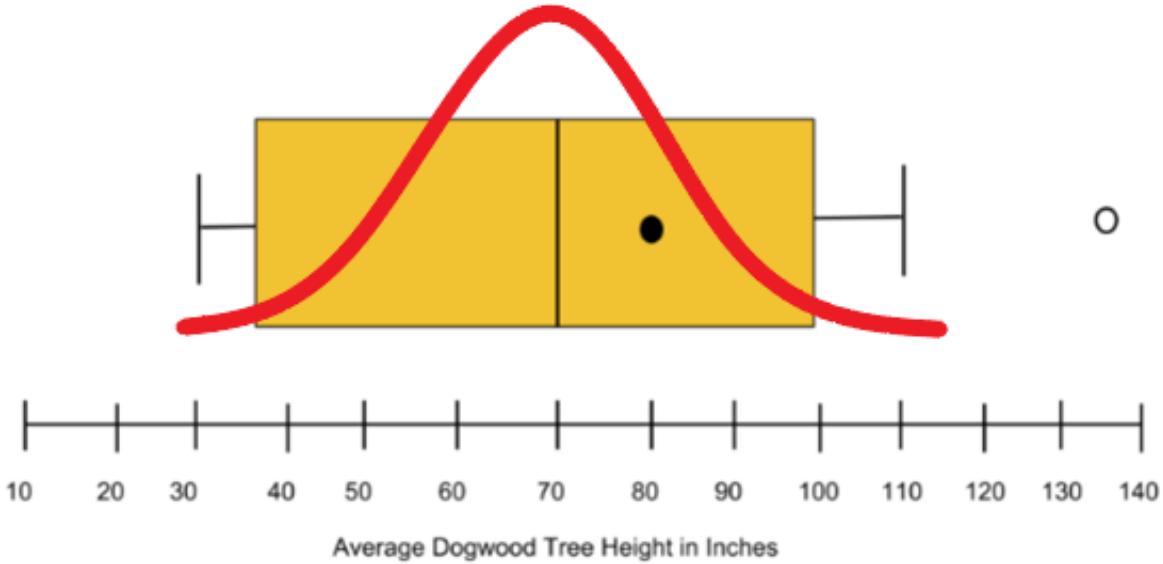
(Graphs Source)



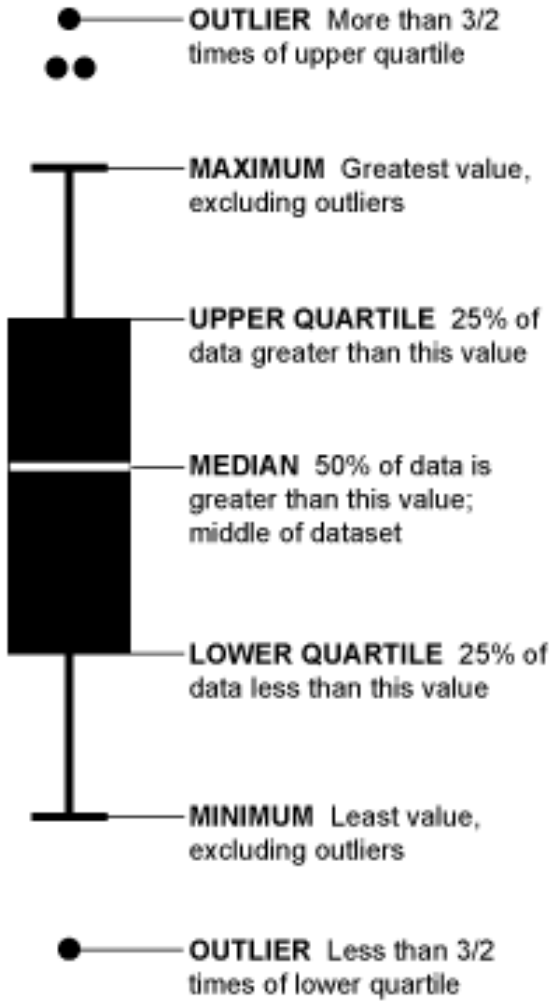
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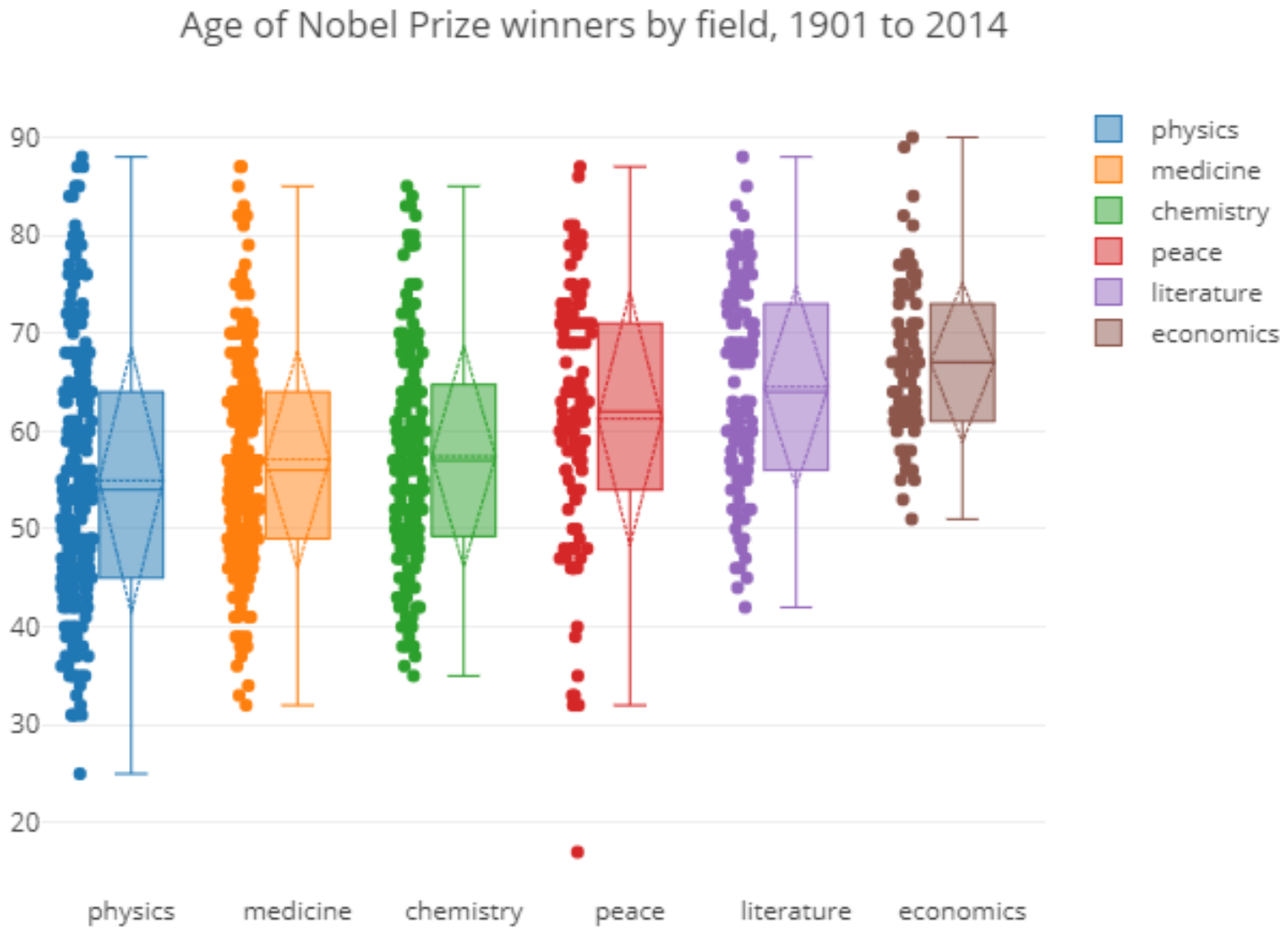


(Graphs Source)



Boxplots

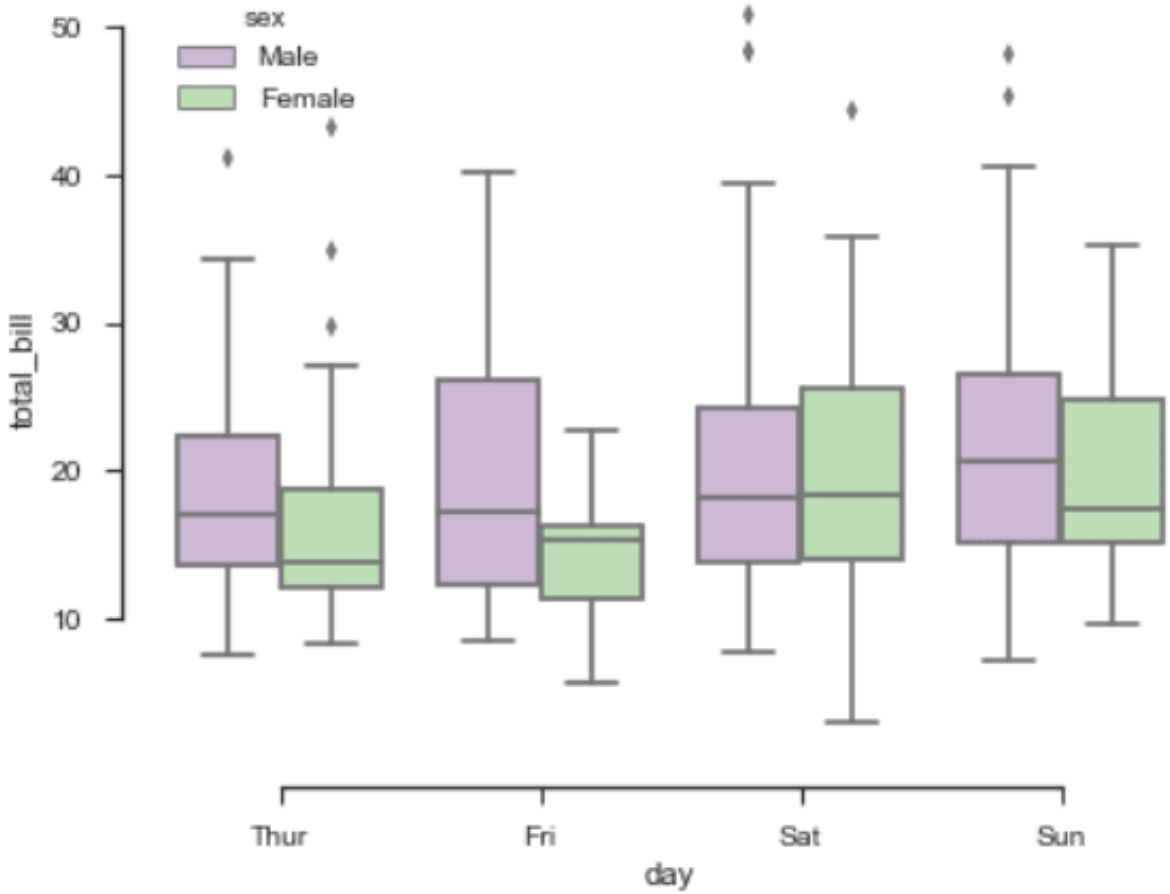
Boxplots are typically used to compare distributions!



(Graphs Source)

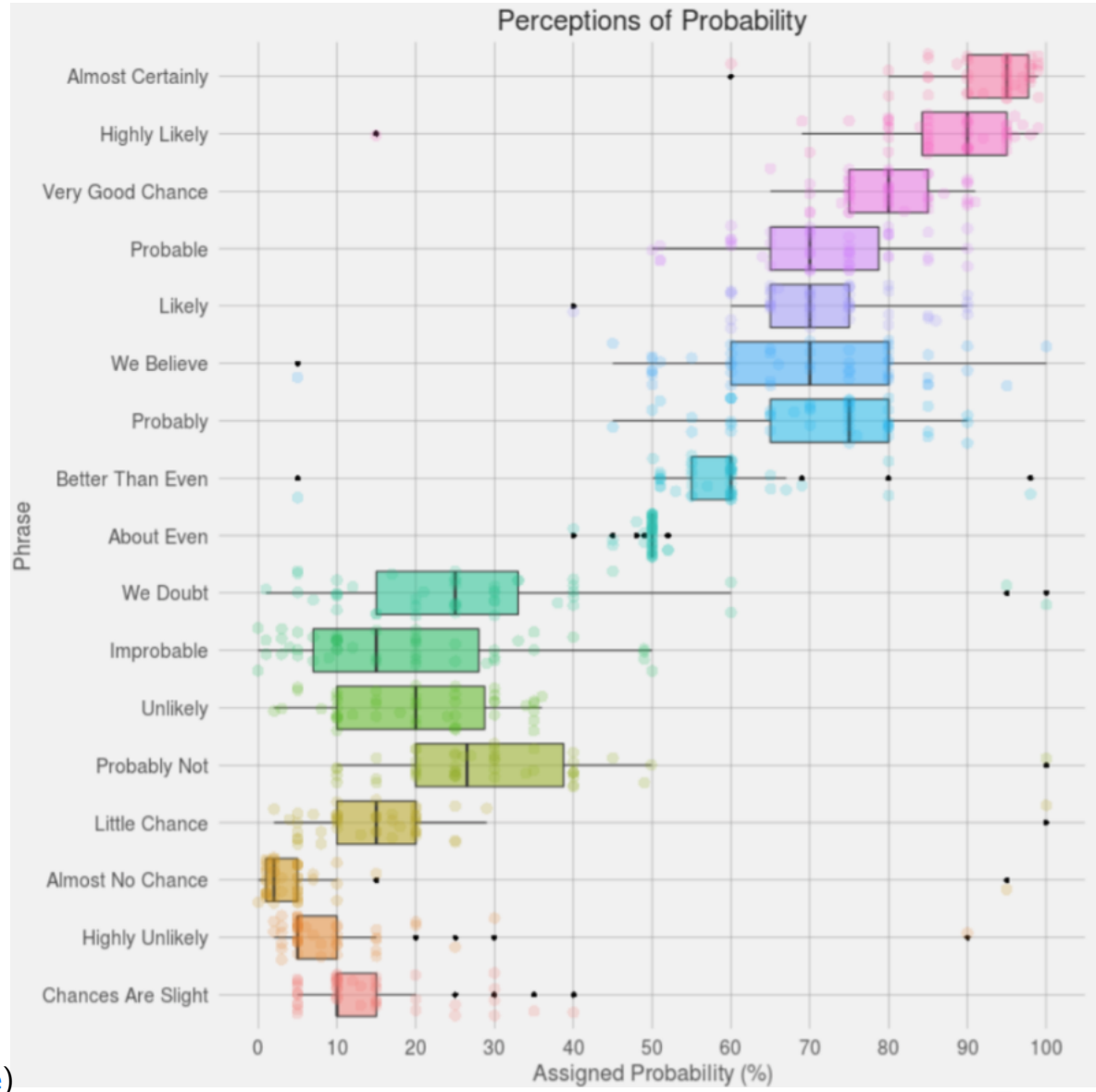
Boxplots

Boxplots are most often used to compare group distributions! This graph looks at how much of the total bill men and women pay on a given date on common date nights.



(Graphs Source)

Boxplots



(Graphs Source)

Class Discussion 2: History of Graphs

Class Discussion: We will watch the following YouTube Video and discuss.

Florence Nightingale: Joy of Stats (3/6) (3:41)

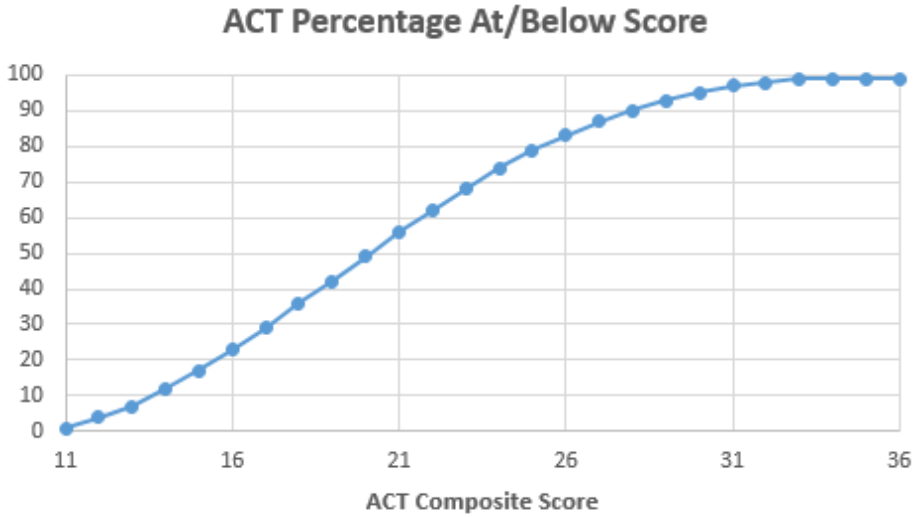
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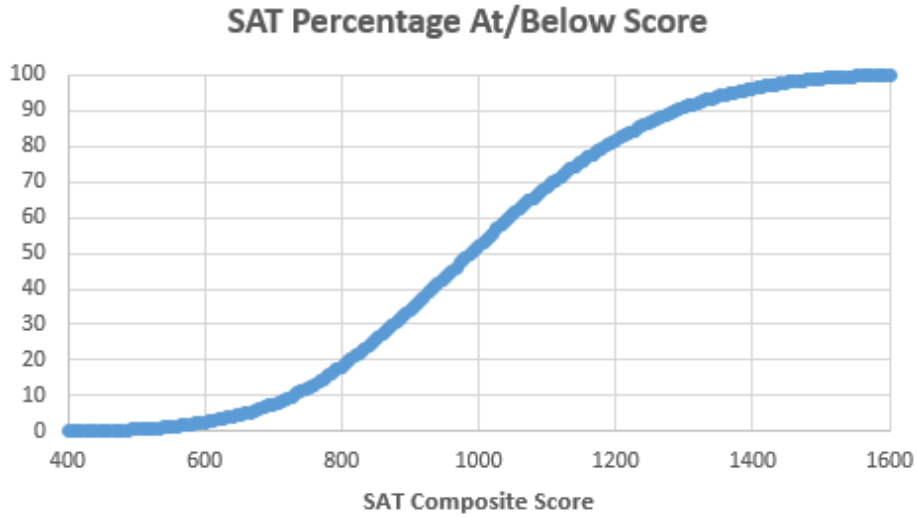
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Percentiles

Percentiles

A **percentile** is the percentage of subjects who have scores at or below that value. A **percentile score** is the score having a stated percentage of subjects 'at or below' that value.





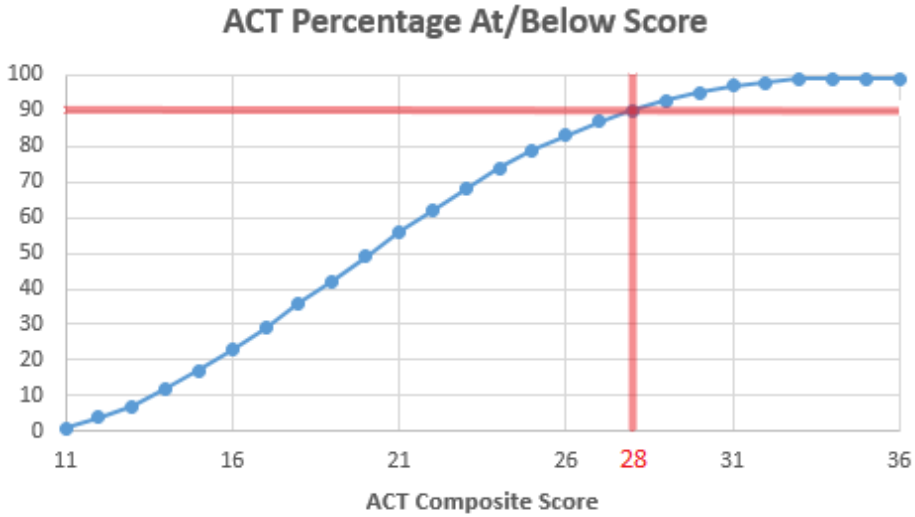
The 90% percentile score on the ACT is a 28. So, 90% of students taking the ACT will score a 28 or below. Alternatively,

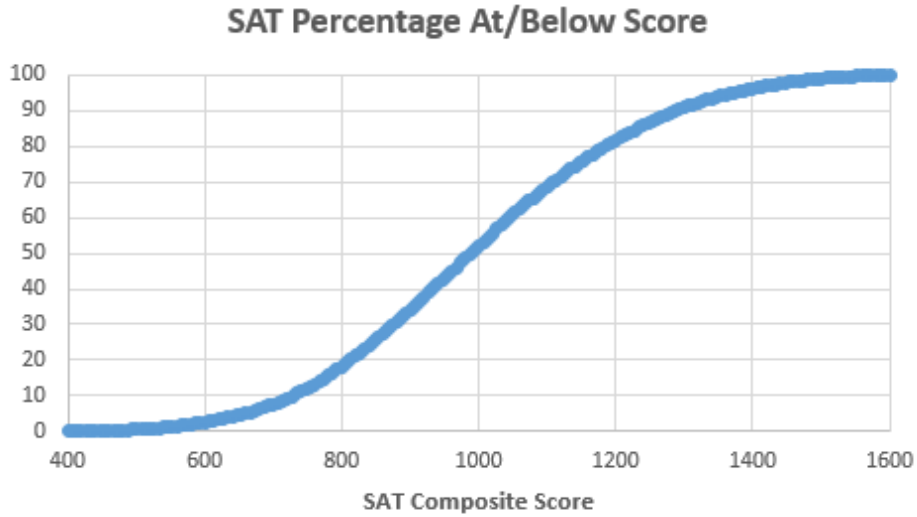
- A student scoring a 28 or higher on the ACT is at the 90th percentile.
- A student scoring a 28 on the ACT is in the top 10% of ACT scores.

What is the (approximate) 90th percentile for the SAT?

Percentiles

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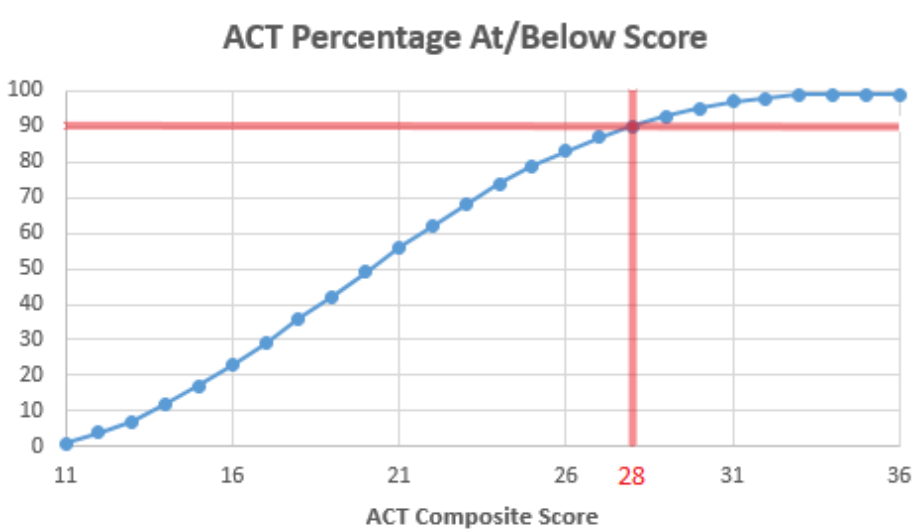


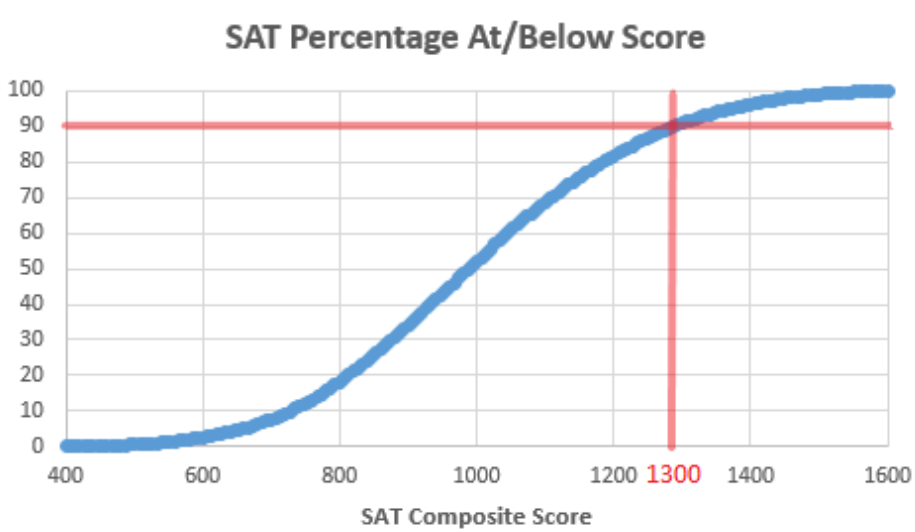
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Percentiles

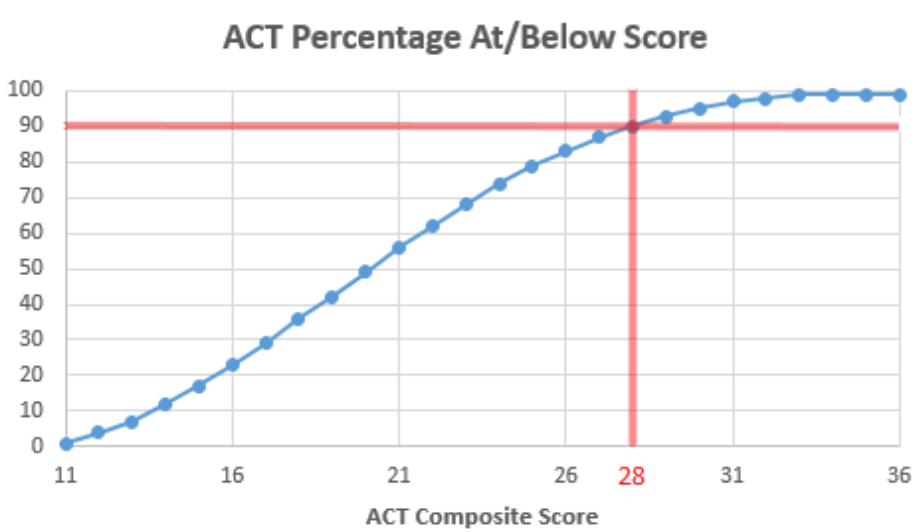


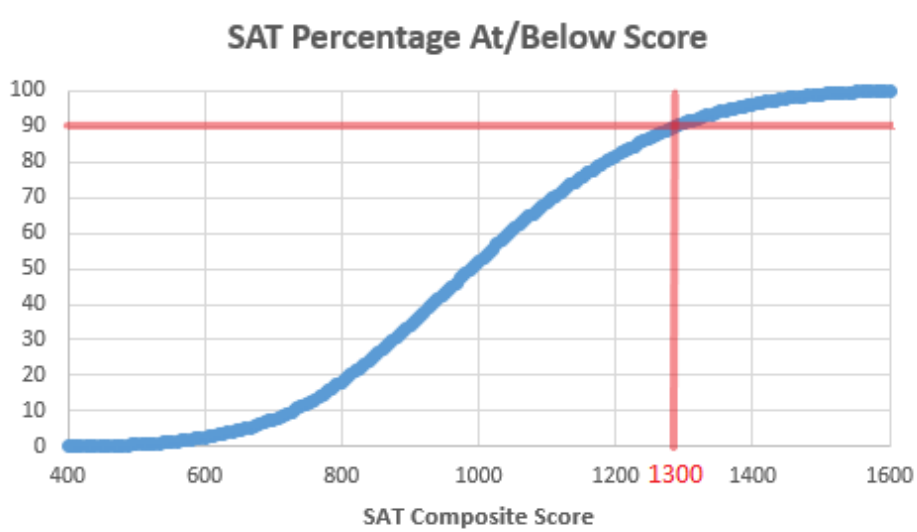


From these charts, you see that a 1300 on the SAT is at the 90th percentile and a 28 on the ACT is also at the 90th percentile.

- If the students taking the ACT and the SAT are similar, then there is good reason to treat a 28 ACT score the same as a 1300 SAT score!

Percentiles





To be statistically literate, you need to realize that percentiles are NOT the same as percentages!

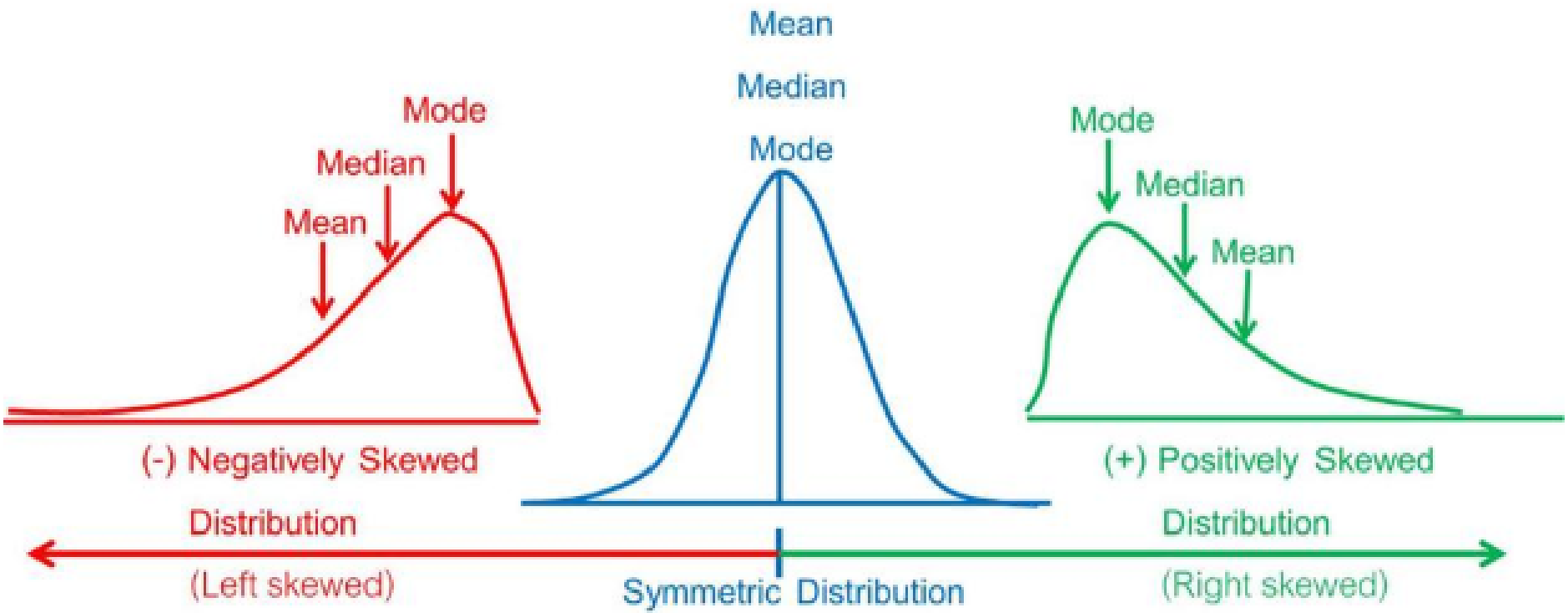
- The fact that a student's score was at the 90th percentile on the SAT does not mean that they got a 90% on that test!
  - On the SAT, $1300/1600 = 81\%$!!
  - On the ACT, $28/36 = 78\%$!!
- The percentile score shows how a student's own score stands in relation to ALL the scores on the test.

Measures of Center and Spread

Most Common Measures of Location and Spread

Measures of Location	Measures of Spread
Mean	**Standard Deviation**
◦ Most frequently used ◦ $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ ◦ $\bar{X}$ estimates the unknown population mean μ ◦ *Highly influenced by outliers!*	◦ Most frequently used ◦ $s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}$ ◦ s estimates the unknown population standard deviation σ ◦ *Highly influenced by outliers!*
Median	**Interquartile Range**
◦ Use instead of mean when the data is skewed ◦ Middle Value ◦ *NOT influenced by outliers!*	◦ Use instead of SD when the data is skewed ◦ (25th percentile, 75th percentile) ◦ *NOT influenced by outliers!*

Measurements and Skewness



Why measures of spread?

Much statistical evidence compares measures of center of different distributions. Consider the following claims:

- The mean IQ is 10% less for high school graduates than for college graduates.
- The modal weight is 5 pounds more for this group than for that group.
- The median house price is 30% more in this neighborhood than in that neighborhood.

But these comparisons don't tell you whether the comparison is small or large – compared to the full distribution!

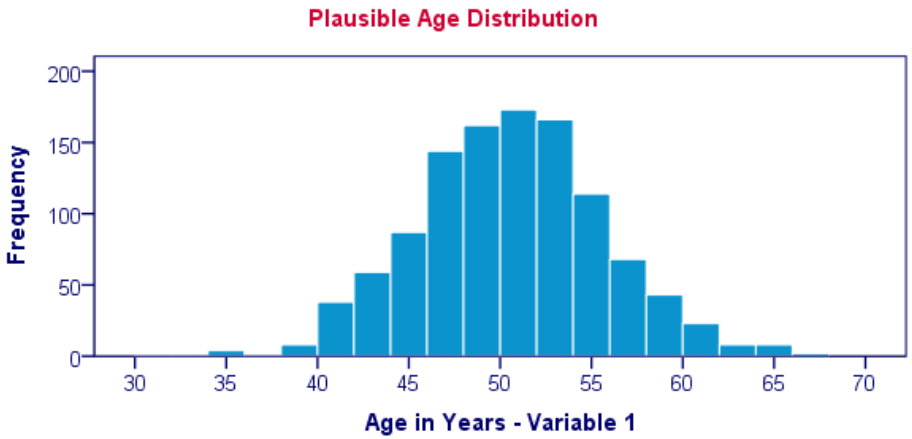
- For houses, a 30% difference could be as low as \$30,000 in a low cost neighborhood
- or as high as \$300,000 in a very high cost neighborhood!

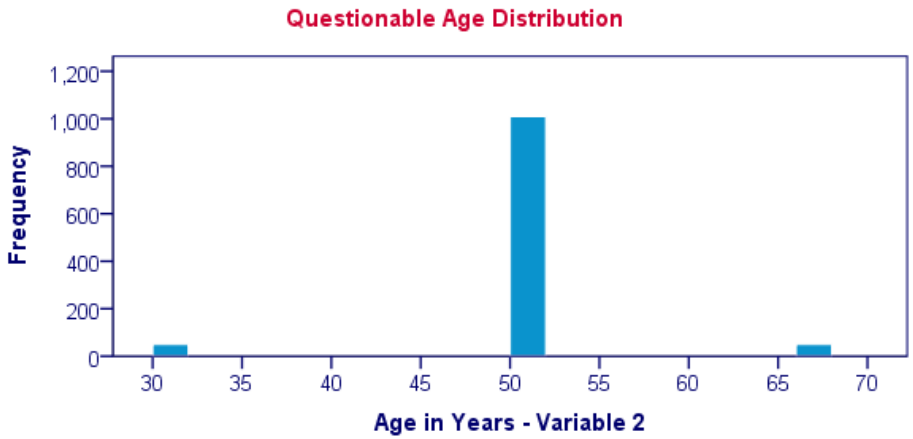
Graphs Versus Summary Statistics

The statistics sometimes don't tell the whole story!

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Age in Years - Variable 1	1110	31	66	50.1	5.1
Age in Years - Variable 2	1110	31	66	49.9	5.3





(Table and Graph Source)

Questions?



Outline

1. Understanding measurements
   - Data Types
   - Exploratory Data Analysis
2. EDA: Graphs
   - Distributions
     - Common Distribution Graphs
     - For Categorical Data
     - For Continuous Data
3. EDA: Statistics
   - Percentiles
   - Measures of Center and Spread
4. References

Sources

A lot of this lecture was taken verbatim from :

- Statistical Literacy 2011: Seeing The Story Behind the Statistics by Milo Schield
  - ISBN: 978-1450787413

Suggested YouTube Videos

Data Types

1. Types of Data: Nominal, Ordinal, Interval/Ratio - Statistics Help (6:19)
2. Nominal, ordinal, interval, and ratio data: How to Remember the differences (11:03)
3. Scales of Measurement - Nominal, Ordinal, Interval, Ratio (Part 1) - Intro Stats (5:51)
4. Scales of Measurement - Nominal, Ordinal, Interval, Ratio (Part 2) - Intro Stats (5:13)

Suggested YouTube Videos

Summary Statistics

1. Mean, Median, and Mode: Measures of Central Tendency: Crash Course Statistics #3 (11:22)
2. Measures of Spread: Crash Course Statistics #4 (11:46)
3. Understanding Summary statistics: Mean, Median, Mode (5:13)